Syllabus Covered:

- Introduction to Computer Networks
- OSI Reference Model (7 Layers)
- TCP/IP Model (4/5 Layers)

1. Introduction to Computer Networks

Definition

A Computer Network is a collection of autonomous computers and peripheral devices interconnected by communication channels that allow users to share resources, exchange data, and communicate electronically. It involves the use of hardware (routers, cables) and software (protocols) to facilitate seamless data transfer.

Detailed Explanation

Computer networks have evolved from simple point-to-point connections to the vast global internet we use today. The primary objective is **Resource Sharing**, which includes sharing printers, scanners, or high-end processing units. Another critical aspect is **Information Sharing**, enabling instant access to databases and file systems across the globe.

Reliability is a major factor in networking. By having multiple paths between devices, if one link fails, another can take over, ensuring high availability. Networks are classified based on their scale: LAN (Local Area Network) for small offices, MAN (Metropolitan Area Network) for cities, and WAN (Wide Area Network) for countries or continents.

Key Features:

- Fault Tolerance: The ability of the network to stay functional during link failures.
- Scalability: Ease of adding new users/devices without degrading performance.
- Security: Protecting data from unauthorized access through encryption and firewalls.
- Performance: Measured in terms of throughput (data speed) and latency (delay).

Applications

- E-commerce and Online Banking.
- Streaming services (Netflix, YouTube).
- Cloud Computing (AWS, Azure).
- Social Media and Instant Messaging.

> 👉 ***Hinglish Explanation:*** *Computer Network matlab bahut saare computers ko aapas mein connect karna taaki hum data aur resources (jaise printer) share kar sakein. Isse hum communication fast kar sakte hain aur data ko secure rakh sakte hain.*

2. OSI Reference Model

Definition

The Open Systems Interconnection (OSI) model is a conceptual framework developed by the ISO in 1984. It characterizes and standardizes the communication functions of a computing system into seven distinct layers without regard to its underlying internal structure and technology.

Detailed Explanation

The OSI model is a "logical" model that helps developers and network professionals understand how data travels from a software application on one computer to a software application on another. It divides the complex process of networking into 7 manageable layers. Each layer serves the layer above it and is served by the layer below it.

The top three layers (Application, Presentation, Session) are often referred to as the **Upper Layers** or Software Layers, primarily dealing with user interface and data formatting. The bottom four layers (Transport, Network, Data Link, Physical) are the **Lower Layers** or Hardware Layers, dealing with the actual movement of bits and packets across the network.

The Seven Layers (Bottom to Top):

1. **Physical Layer:** Deals with bits, cables, and electrical signals.
2. **Data Link Layer:** Responsible for node-to-node delivery and framing. (MAC Addresses)
3. **Network Layer:** Responsible for routing and logical addressing. (IP Addresses)
4. **Transport Layer:** Ensures end-to-end delivery and error recovery. (TCP/UDP)
5. **Session Layer:** Manages sessions between applications (Establish, Maintain, Terminate).
6. **Presentation Layer:** Deals with data encryption, compression, and translation.
7. **Application Layer:** Provides network services directly to the user (HTTP, FTP).

```
[ Application ]  <- Layer 7 (Data)
[ Presentation ] <- Layer 6 (Data)
[ Session ]      <- Layer 5 (Data)
[ Transport ]    <- Layer 4 (Segments)
[ Network ]      <- Layer 3 (Packets)
[ Data Link ]    <- Layer 2 (Frames)
[ Physical ]     <- Layer 1 (Bits)
```

Advantages:

- Standardization of network components.
- Simplifies troubleshooting by isolating problems to specific layers.
- Allows different vendors' products to communicate.

> 👉 ***Hinglish Explanation:*** *OSI model ek 7-layer ka structure hai jo batata hai ki data ek computer se doosre computer tak kaise jayega. Har layer ka apna fix kaam hota hai—jaise Network layer rasta dikhati hai aur Physical layer wires se bits bhejti hai.*

3. TCP/IP Model

Definition

The Transmission Control Protocol/Internet Protocol (TCP/IP) model is a concise, four-layer hierarchical protocol suite developed by the DoD (Department of Defense). It is the actual practical model used for the modern Internet.

Detailed Explanation

Unlike the OSI model which is theoretical, the TCP/IP model is functional. It was designed to be flexible and robust, allowing data to reach its destination even if some intermediate nodes fail. It combines some of the OSI layers to create a more streamlined process. For example, the OSI Application, Presentation, and Session layers are merged into a single Application layer in TCP/IP.

The core of this model is the **Internet Layer** (IP) and the **Transport Layer** (TCP). IP handles the addressing and routing of packets, while TCP ensures that those packets are delivered reliably and in the correct order. Because of its efficiency, it replaced other protocols to become the standard for global networking.

Application	User interface & Apps	HTTP, SMTP, DNS
Transport	End-to-end communication	TCP, UDP
Internet	Logical addressing & Routing	IP, ICMP, ARP
Network Access	Physical transmission	Ethernet, Wi-Fi

Key Differences from OSI:

- TCP/IP is 4 layers (standard) vs OSI's 7 layers.
- TCP/IP is protocol-dependent; OSI is protocol-independent.
- TCP/IP combined layers make it faster and more practical for implementation.

> 👉 ***Hinglish Explanation:*** *TCP/IP model practical model hai jo hum aaj Internet pe use karte hain. Ismein sirf 4 layers hoti hain. Yeh OSI se purana hai par zyada popular hai kyunki yeh seedha implementation pe focus karta hai.*

🔁 Short Revision Notes

- **Network:** Interconnected computers for sharing.
- **OSI:** 7 Layers (APSTNDP - All People Seem To Need Data Processing).
- **TCP/IP:** 4 Layers (Application, Transport, Internet, Network Access).
- **MAC vs IP:** MAC is physical (Layer 2), IP is logical (Layer 3).

📝 Important Questions (6 Marks Each)

1. Define Computer Networks and explain different types of network topologies in detail.
2. Explain the 7 layers of the OSI Reference Model with their specific functions and protocols.
3. Compare and contrast the OSI model and TCP/IP model with a neat diagram.
4. Explain the functions of the Network Layer and Transport Layer in the TCP/IP suite.

5. What is the need for Layered Architecture in Computer Networks? Discuss the advantages.

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